

## Lecture 3

### Linear Maps: Kernel, Image, Rank–Nullity

We have the objects of linear algebra—vector spaces—and their coordinate systems—bases. Now we study the *maps between* vector spaces that respect this structure. A *linear map* is a function that commutes with addition and scalar multiplication; equivalently, it preserves linear combinations, so it preserves superposition.

This is also why linear maps are the right notion of a “transformation” in physics. In quantum mechanics linear operators do three different jobs. One kind of operator encodes a measurable quantity. A second describes how a closed system evolves in time. A third describes a symmetry. Measurement itself does something that is not a linear operation: it selects the part of a state belonging to one possible outcome and replaces the state by that part. We will define all of this in Lectures 7–10, and nothing in the present lecture depends on it.

Two subspaces measure how a linear map behaves: its *kernel* (what it sends to zero) and its *image* (what it can produce). The central result of the lecture, the rank–nullity theorem, ties their dimensions together in a single equation  $\dim V = \dim \ker T + \dim \operatorname{im} T$ , and from it we will read off when linear systems are solvable and when operators are invertible.

#### Learning objectives

By the end of this lecture you will be able to:

- ▶ Check linearity and build a map freely from its values on a basis.
- ▶ Compute the kernel and image and read off injectivity and surjectivity.
- ▶ Apply the rank–nullity theorem and its dimension obstructions.
- ▶ Recognise the quotient space and First Isomorphism Theorem (self-study).

## 1 Linear maps and their examples

**Definition 3.1 (Linear map).** Let  $V, W$  be vector spaces over the same field  $\mathbb{F}$ . A function  $T: V \rightarrow W$  is a *linear map* (or *linear transformation*) if

$$T(u + v) = Tu + Tv \quad \text{and} \quad T(av) = aTv \quad \text{for all } u, v \in V, a \in \mathbb{F}.$$

The set of all linear maps  $V \rightarrow W$  is written  $\mathcal{L}(V, W)$ . A linear map from  $V$  to itself is an *operator*, and we write  $\mathcal{L}(V)$  for  $\mathcal{L}(V, V)$ .

Applying the two rules repeatedly, a linear map sends 0 to 0 (take  $a = 0$ ) and preserves every linear combination:

$$T(a_1 v_1 + \cdots + a_m v_m) = a_1 T v_1 + \cdots + a_m T v_m. \quad (3.1)$$

This single identity is the reason linear maps are ubiquitous.

**Example 3.2 (A catalogue of linear maps).** Each of the following is linear, as the two rules readily confirm.

- The *zero map*  $0: V \rightarrow W$  and the *identity*  $I: V \rightarrow V, Iv = v$ .
- *Matrix maps:* for  $A \in M_{m \times n}(\mathbb{F})$ , the map  $x \mapsto Ax$  from  $\mathbb{F}^n$  to  $\mathbb{F}^m$ . In Lecture 4 we will see that, in coordinates, *every* linear map looks like this.

- *Differentiation*  $D: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ ,  $Dp = p'$ , and more generally on spaces of differentiable functions.
- *Multiplication*  $M: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ ,  $(Mp)(t) = t p(t)$ .
- The *backward shift* on  $\mathbb{F}^\infty$ ,  $T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$ .
- *Rotation* of  $\mathbb{R}^2$  by an angle  $\theta$ , and the *evaluation* functional  $\mathcal{P}(\mathbb{F}) \rightarrow \mathbb{F}$ ,  $p \mapsto p(c)$ . ◀

**Example 3.3 (Testing linearity).** The map  $S(x, y) = (2x - y, x)$  on  $\mathbb{R}^2$  is linear: each output coordinate is a linear combination of  $x, y$  with no constant term. The map  $T(x, y) = (x+1, y)$  is *not* linear—it fails  $T0 = 0$ , since  $T(0, 0) = (1, 0) \neq 0$ . A constant term anywhere is fatal to linearity. ◀

**Example 3.4 (Why the catalogue works).** Differentiation is linear because of the single identity  $(af + bg)' = af' + bg'$ . The same “apply the rule to each piece” pattern—valid for the backward shift, integration, multiplication by a fixed function, and matrix multiplication—is what makes every entry of Example 3.2 linear. Linearity is rarely hard to check; the hard question is what a given linear map *does*, which is the rest of the lecture. ◀

## 2 A linear map is determined by a basis

A linear map cannot be chosen freely at every point: (3.1) ties its values together. In fact its values on a single basis fix it completely, and those values may be assigned arbitrarily.

**Theorem 3.5 (Maps are determined by a basis).** Let  $v_1, \dots, v_n$  be a basis of  $V$  and let  $w_1, \dots, w_n$  be *any* vectors in  $W$ . Then there is a unique linear map  $T: V \rightarrow W$  with  $Tv_k = w_k$  for each  $k$ .

*Proof. Existence.* Each  $v \in V$  has a unique expression  $v = \sum_k a_k v_k$  (Lecture 2); define  $Tv = \sum_k a_k w_k$ . Linearity follows because coordinates add and scale: if  $v = \sum a_k v_k$  and  $u = \sum b_k v_k$  then  $v + u = \sum (a_k + b_k) v_k$ , so  $T(v + u) = \sum (a_k + b_k) w_k = Tv + Tu$ , and similarly  $T(cv) = cTv$ . Clearly  $Tv_k = w_k$ .

*Uniqueness.* If  $S$  is linear with  $Sv_k = w_k$ , then for any  $v = \sum a_k v_k$ , (3.1) gives  $Sv = \sum a_k w_k = Tv$ . So  $S = T$ . ■

**Corollary 3.6 (Agreement on a basis).** If two linear maps  $S, T \in \mathcal{L}(V, W)$  satisfy  $Sv_k = Tv_k$  on a basis  $v_1, \dots, v_n$  of  $V$ , then  $S = T$ .

*Proof.* Both are linear maps with the same values on the basis, so both equal the unique map of Theorem 3.5. ■

**Example 3.7 (Building a map from its values on a basis).** Define  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  by its action on the basis  $(1, 1), (1, -1)$ :

$$T(1, 1) = (2, 0), \quad T(1, -1) = (0, 2).$$

To evaluate  $T$  on a general  $(x, y)$ , first find coordinates: writing  $(x, y) = a(1, 1) + b(1, -1)$  gives  $a = \frac{x+y}{2}$ ,  $b = \frac{x-y}{2}$ . Then

$$T(x, y) = a(2, 0) + b(0, 2) = (2a, 2b) = (x + y, x - y).$$

Theorem 3.5 guarantees this is the one and only linear map with the prescribed values. ◀

### 3 The algebra of linear maps

Linear maps can be added, scaled, and composed.

**Definition 3.8 (Operations on linear maps).** For  $S, T \in \mathcal{L}(V, W)$  and  $a \in \mathbb{F}$ , define  $S + T$  and  $aS$  pointwise:  $(S + T)v = Sv + Tv$  and  $(aS)v = a(Sv)$ . For  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$ , the *product* (composition)  $ST: U \rightarrow W$  is  $(ST)u = S(Tu)$ .

Addition and scaling are defined for maps between the same pair of spaces. The product  $ST$  is defined only when the codomain of  $T$  is the domain of  $S$ . So the proposition states, for every map, between which spaces it acts.

**Proposition 3.9 (The algebra of linear maps).**

- (1) With these operations,  $\mathcal{L}(V, W)$  is a vector space.
- (2) The product of linear maps  $T \in \mathcal{L}(U, V)$  and  $S \in \mathcal{L}(V, W)$  is linear too:  $ST \in \mathcal{L}(U, W)$ .
- (3) The product is associative, distributes over addition, and has the identity maps as units. That is, for all  $T, T_1, T_2 \in \mathcal{L}(U, V)$ ,  $S, S_1, S_2 \in \mathcal{L}(V, W)$  and  $R \in \mathcal{L}(W, X)$ ,

$$(RS)T = R(ST), \quad S(T_1 + T_2) = ST_1 + ST_2, \quad (S_1 + S_2)T = S_1T + S_2T,$$

and, for the identity maps  $I_V$  and  $I_W$  of  $V$  and  $W$ ,

$$I_W S = S I_V = S.$$

*Proof.* (1)  $\mathcal{L}(V, W)$  is a subspace of the space  $W^V$  of all functions  $V \rightarrow W$ : the zero map is linear, and if  $S, T$  are linear and  $a \in \mathbb{F}$  then so are  $S + T$  and  $aS$ , since, for example,

$$(S + T)(u + v) = S(u + v) + T(u + v) = Su + Sv + Tu + Tv = (S + T)u + (S + T)v.$$

(2) Apply the linearity of  $T$ , then the linearity of  $S$ :

$$(ST)(u + u') = S(Tu + Tu') = (ST)u + (ST)u' \quad \text{and} \quad (ST)(au) = S(aTu) = a(ST)u.$$

(3) Each identity is checked by applying both sides to an arbitrary vector  $u \in U$ . For example, by the linearity of  $S$ ,

$$(S(T_1 + T_2))u = S(T_1u + T_2u) = S(T_1u) + S(T_2u) = (ST_1 + ST_2)u.$$

The remaining identities are checked in the same way. ■

For operators, that is when  $U = V = W = X$ , all these products are defined. So operators on  $V$  can be added, scaled and multiplied with one another, and the result is again an element of  $\mathcal{L}(V)$ . Products of operators, however, **need not commute**—and the failure to commute carries physical content.

**Example 3.10 (A commutator: the canonical commutation relation).** On  $\mathcal{P}(\mathbb{R})$  let  $D$  be differentiation and  $M$  multiplication by  $t$ . For any  $p$ ,

$$(DM)p = (t p(t))' = p(t) + t p'(t), \quad (MD)p = t p'(t),$$

so  $(DM - MD)p = p$  for every  $p$ ; that is, the *commutator*  $[D, M] := DM - MD$  equals the identity:

$$[D, M] = I. \tag{3.2}$$

In quantum mechanics the position and momentum operators obey the analogous relation  $[\hat{x}, \hat{p}] = i\hbar I$ . Non-commuting operators are the rule, not the exception, and (3.2) is your first sighting of why. ◀

**Example 3.11 (Concrete non-commuting operators).** The phenomenon is visible already with  $2 \times 2$  matrices. Let  $S, T \in \mathcal{L}(\mathbb{R}^2)$  be the matrix maps with

$$S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$ST = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad TS = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad [S, T] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0.$$

( $S$  and  $T$  are the raising and lowering operators of a two-level system.)

**Remark 3.12.** Once bases of finite-dimensional  $V$  and  $W$  are fixed, addition, scaling, and composition of linear maps become exactly addition, scaling, and multiplication of their matrices—the subject of Lecture 4. In that finite setting one can show  $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$ ; the proof is cleanest in matrix language, so we defer it. No corresponding product formula is asserted here for arbitrary infinite-dimensional algebraic map spaces.

## 4 Kernel and injectivity

**Definition 3.13 (Kernel (null space)).** The *kernel* (or *null space*) of  $T \in \mathcal{L}(V, W)$  is

$$\ker T = \{v \in V : Tv = 0\} \subseteq V.$$

**Proposition 3.14 (The kernel is a subspace).**  $\ker T$  is a subspace of  $V$ .

*Proof.*  $T0 = 0$ , so  $0 \in \ker T$ . If  $u, v \in \ker T$  and  $a \in \mathbb{F}$  then  $T(u + v) = Tu + Tv = 0$  and  $T(av) = aTv = 0$ , so  $u + v$  and  $av$  lie in  $\ker T$ . The subspace test is met. ■

**Example 3.15 (Kernels).** The kernel of differentiation  $D$  on  $\mathcal{P}(\mathbb{R})$  is the constants (the functions with zero derivative). The kernel of multiplication by  $t$  is  $\{0\}$ . The kernel of the backward shift on  $\mathbb{F}^\infty$  is  $\{(a, 0, 0, \dots) : a \in \mathbb{F}\}$ . And the kernel of a matrix map  $x \mapsto Ax$  is exactly the solution set of the homogeneous system  $Ax = 0$ —the null space you met when row-reducing  $A$ .

**Example 3.16 (Computing a kernel).** Let  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  be  $Tx = Ax$  with  $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{pmatrix}$ . The kernel is the solution set of  $Ax = 0$ . Subtracting twice the first row from the second gives  $(0, 0, 3)$ , so  $x_3 = 0$ ; the first row then reads  $x_1 + 2x_2 = 0$ , i.e.  $x_1 = -2x_2$ , with  $x_2$  free. Hence

$$\ker T = \text{span}((-2, 1, 0)), \quad \dim \ker T = 1.$$

We return to this map in [Example 3.22](#) to see its image.

**Definition 3.17 (Injective).**  $T: V \rightarrow W$  is *injective* (one-to-one) if  $Tu = Tv$  implies  $u = v$ .

**Theorem 3.18 (Injectivity is a kernel condition).**  $T \in \mathcal{L}(V, W)$  is injective if and only if  $\ker T = \{0\}$ .

*Proof.* If  $T$  is injective and  $v \in \ker T$ , then  $Tv = 0 = T0$  forces  $v = 0$ , so  $\ker T = \{0\}$ . Conversely, if  $\ker T = \{0\}$  and  $Tu = Tv$ , then  $T(u - v) = Tu - Tv = 0$ , so  $u - v \in \ker T = \{0\}$ , giving  $u = v$ . Thus  $T$  is injective. ■

## 5 Image and surjectivity

**Definition 3.19 (Image; rank).** The *image* of  $T \in \mathcal{L}(V, W)$  is  $\text{im } T = \{Tv : v \in V\} \subseteq W$ . When  $\text{im } T$  is finite-dimensional—in particular whenever  $V$  is, since the images of a basis of  $V$  span  $\text{im } T$ —its dimension  $\dim \text{im } T$  is the *rank* of  $T$ . The map  $T$  is *surjective* (onto) if  $\text{im } T = W$ .

**Proposition 3.20 (The image is a subspace).**  $\text{im } T$  is a subspace of  $W$ .

*Proof.*  $0 = T0 \in \text{im } T$ . If  $w_1 = Tv_1$  and  $w_2 = Tv_2$  then  $w_1 + w_2 = T(v_1 + v_2) \in \text{im } T$ , and  $aw_1 = T(av_1) \in \text{im } T$ . The subspace test holds. ■

**Example 3.21 (Images, and dependence on the target).** The image of a matrix map  $x \mapsto Ax$  is the span of the columns of  $A$ —the column space. Differentiation  $D: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$  is surjective, since every polynomial has an antiderivative. But surjectivity depends on the chosen target:  $D: \mathcal{P}_5(\mathbb{R}) \rightarrow \mathcal{P}_5(\mathbb{R})$  is *not* surjective ( $t^5$  is not a derivative of anything in  $\mathcal{P}_5$ ), whereas  $D: \mathcal{P}_5(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$  *is*. ◀

**Example 3.22 (Computing a image).** For the map  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$  of [Example 3.16](#), the image is the column space of  $A$ . Its columns are  $(1, 2)$ ,  $(2, 4)$ ,  $(-1, 1)$ ; since  $(2, 4) = 2(1, 2)$ , two independent columns remain, so

$$\text{im } T = \text{span}((1, 2), (-1, 1)) = \mathbb{R}^2, \quad \text{rank } T = 2.$$

Thus  $T$  is surjective, and the dimensions balance:  $3 = \dim \ker T + \text{rank } T = 1 + 2$ , a preview of the theorem below. ◀

A linear map carries spanning lists to spanning lists, and—when injective—it preserves independence.

**Proposition 3.23 (Action on lists).** Let  $T \in \mathcal{L}(V, W)$  and  $v_1, \dots, v_n \in V$ .

1. If  $v_1, \dots, v_n$  spans  $V$ , then  $Tv_1, \dots, Tv_n$  spans  $\text{im } T$ .
2. If  $T$  is injective and  $v_1, \dots, v_n$  is linearly independent, then  $Tv_1, \dots, Tv_n$  is linearly independent.

*Proof.* (1) Every element of  $\text{im } T$  is  $Tv$  for some  $v = \sum_k a_k v_k$ , and then  $Tv = \sum_k a_k Tv_k$  lies in  $\text{span}(Tv_1, \dots, Tv_n)$ . (2) If  $\sum_k c_k Tv_k = 0$  then  $T(\sum_k c_k v_k) = 0$ , so  $\sum_k c_k v_k$  is in  $\ker T = \{0\}$  by injectivity; independence of the  $v_k$  then forces every  $c_k = 0$ . ■

## 6 The Fundamental Theorem of Linear Maps

The kernel records what  $T$  destroys; the image records what it builds. The next theorem—the central result of the lecture—says their dimensions exactly account for the dimension of the domain ([Figure 1](#)).

**Theorem 3.24 (Rank–nullity).** Suppose  $V$  is finite-dimensional and  $T \in \mathcal{L}(V, W)$ . Then  $\text{im } T$  is finite-dimensional and

$$\dim V = \dim \ker T + \dim \text{im } T.$$

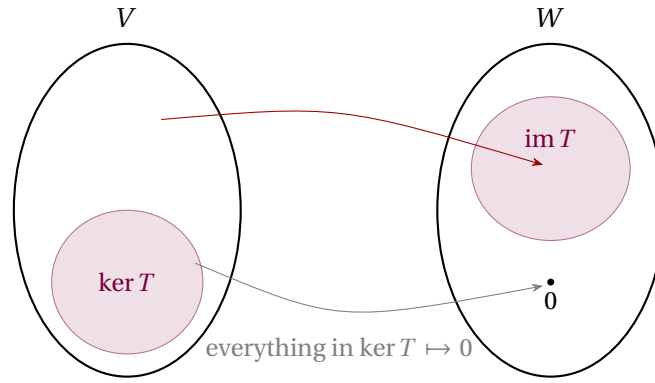


Figure 1: A linear map collapses its kernel to 0 and carries a complement of the kernel one-to-one onto its image; the dimensions balance as  $\dim V = \dim \ker T + \dim \operatorname{im} T$ .

*Proof.* Let  $u_1, \dots, u_m$  be a basis of  $\ker T$ , so  $\dim \ker T = m$ . Extend it (Lecture 2) to a basis  $u_1, \dots, u_m, v_1, \dots, v_n$  of  $V$ , so  $\dim V = m + n$ . We show  $Tv_1, \dots, Tv_n$  is a basis of  $\operatorname{im} T$ , which gives  $\dim \operatorname{im} T = n$  and finishes the count.

*Spanning.* Any  $v \in V$  is  $v = \sum_i a_i u_i + \sum_j b_j v_j$ ; applying  $T$  and using  $Tu_i = 0$  leaves  $Tv = \sum_j b_j Tv_j$ . Hence  $Tv_1, \dots, Tv_n$  spans  $\operatorname{im} T$  (and  $\operatorname{im} T$  is finite-dimensional).

*Independence.* Suppose  $\sum_j c_j Tv_j = 0$ . Then  $T(\sum_j c_j v_j) = 0$ , so  $\sum_j c_j v_j \in \ker T$  and can be written  $\sum_i d_i u_i$ . Thus

$$\sum_j c_j v_j - \sum_i d_i u_i = 0.$$

Independence of the basis  $u_1, \dots, u_m, v_1, \dots, v_n$  forces all  $c_j = 0$  (and all  $d_i = 0$ ). So  $Tv_1, \dots, Tv_n$  is independent, hence a basis of  $\operatorname{im} T$ . ■

**Example 3.25 (A projection makes the count visible).** Let  $P: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be projection onto the  $x$ -axis,  $P(x, y) = (x, 0)$  (Figure 2). Its kernel is the  $y$ -axis (mapped to 0) and its image is the  $x$ -axis, so  $\dim \ker P = 1$  and  $\dim \operatorname{im} P = 1$ , balancing  $2 = 1 + 1 = \dim \mathbb{R}^2$ . ◀

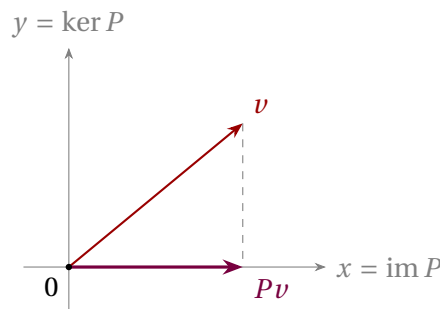


Figure 2: Projection onto the  $x$ -axis: the kernel is the  $y$ -axis, the image is the  $x$ -axis, and  $\dim \ker P + \dim \operatorname{im} P = 1 + 1 = 2$ .

Two consequences follow with no further work, just by comparing dimensions.

**Corollary 3.26 (Dimension obstructions).** Let  $V, W$  be finite-dimensional and  $T \in \mathcal{L}(V, W)$ .

1. If  $\dim V > \dim W$ , then  $T$  is not injective.
2. If  $\dim V < \dim W$ , then  $T$  is not surjective.

*Proof.* (1)  $\dim \ker T = \dim V - \dim \operatorname{im} T \geq \dim V - \dim W > 0$ , so  $\ker T \neq \{0\}$  and  $T$  is not injective by **Theorem 3.18**. (2)  $\dim \operatorname{im} T = \dim V - \dim \ker T \leq \dim V < \dim W$ , so  $\operatorname{im} T \neq W$ . ■

**Example 3.27 (Reading injectivity and surjectivity off dimensions).** No computation is needed to see that  $T(z_1, z_2, z_3, z_4) = (z_1 + z_4, z_2 + z_3, z_1 + z_2)$  from  $\mathbb{R}^4$  to  $\mathbb{R}^3$  has a nonzero kernel: since  $\dim \mathbb{R}^4 > \dim \mathbb{R}^3$ , **Corollary 3.26(1)** forbids injectivity. Dually, no linear map  $\mathbb{R}^2 \rightarrow \mathbb{R}^3$  is surjective, so for any such map the system  $Tx = b$  is unsolvable for some right-hand sides  $b$ . Dimension alone settles both questions. ◀

**Corollary 3.28 (Operators: injective = surjective).** If  $V$  is finite-dimensional and  $T \in \mathcal{L}(V)$ , then  $T$  is injective if and only if it is surjective (and then bijective).

*Proof.* By **Theorem 3.24**,  $\dim \ker T + \dim \operatorname{im} T = \dim V$ . So  $\ker T = \{0\}$  (injective)  $\iff \dim \operatorname{im} T = \dim V \iff \operatorname{im} T = V$  (surjective), using **Theorem 3.18** and that a subspace of full dimension is the whole space. ■

**Corollary 3.28** fails in infinite dimensions: the backward shift on  $\mathbb{F}^\infty$  is surjective but not injective, while the forward shift is injective but not surjective. This contrast is a first warning of the subtleties that Lectures 13–14 confront in Hilbert space.

**Example 3.29 (A rotation is an invertible operator).** Rotation of  $\mathbb{R}^2$  by an angle  $\theta$ ,

$$R_\theta(x, y) = (x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta),$$

is linear, and its kernel is  $\{0\}$  (a rotation carries every nonzero vector to a nonzero vector). By **Corollary 3.28** it is therefore surjective, hence invertible—its inverse is  $R_{-\theta}$ , as geometry demands. Rotations give unitary (here orthogonal) linear symmetries; quantum symmetries more generally may instead be antiunitary and conjugate-linear. ◀

**Example 3.30 (Solving an equation by counting dimensions).** Given  $q \in \mathcal{P}_m(\mathbb{R})$ , does some  $p$  satisfy  $((t^2 + 1)p)'' = q$ ? Define  $T: \mathcal{P}_m(\mathbb{R}) \rightarrow \mathcal{P}_m(\mathbb{R})$  by  $Tp = ((t^2 + 1)p)''$ : multiplying by  $t^2 + 1$  raises the degree by 2 and two derivatives lower it by 2, so  $T$  does map  $\mathcal{P}_m$  into itself. If  $Tp = 0$  then  $(t^2 + 1)p$  has vanishing second derivative, hence is affine; but  $(t^2 + 1)p$  has degree at least 2 unless  $p = 0$ . So  $\ker T = \{0\}$ , i.e.  $T$  is injective, and by **Corollary 3.28** it is surjective. Thus the required  $p$  exists for every  $q$ —injectivity delivered existence with no construction at all. ◀

### Linear systems, revisited

Rank-nullity is the structural statement behind the row-reduction you already know. For  $A \in M_{m \times n}(\mathbb{F})$  and  $T(x) = Ax$ , the rank of  $T$  is the number of pivots and the nullity is the number of free variables, so

$$(\# \text{free variables}) = n - (\# \text{pivots}) = \dim V - \operatorname{rank} T.$$

The system  $Ax = b$  is solvable exactly when  $b \in \operatorname{im} T$  (the column space), and then its full solution set is a single particular solution plus the kernel,  $x_p + \ker T$  (a translate of a subspace). In particular, by **Corollary 3.26(1)**, a homogeneous system with more unknowns than equations always has a nonzero solution.

**Example 3.31 (A homogeneous system with free variables).** The system  $x_1 + x_2 - x_3 + 2x_4 = 0$ ,  $2x_1 + 2x_2 + x_3 + x_4 = 0$  has four unknowns and only two equations, so it must have nonzero

solutions. We write the coefficients into the rows of a matrix and reduce it (there is no need to write the right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 1 & -1 & 2 \\ 2 & 2 & 1 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \implies x_1 = -x_2 - x_4, \quad x_3 = x_4.$$

The pivots are in the first and third columns, so  $x_2$  and  $x_4$  are free. Setting one of them to 1 and the other to 0 reads off the basis, and the solution space is

$$\text{span}((-1, 1, 0, 0), (-1, 0, 1, 1)), \quad \text{of dimension } 2 = 4 - 2.$$

◀

**Example 3.32 (Rank, nullity, and a solution set).** Let  $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$  be  $Tx = Ax$  with

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}.$$

We reduce the matrix  $A$  (there is no need to write a right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \implies x_1 = -2x_2 - x_4, \quad x_3 = -x_4.$$

The pivots are in the first and third columns, so  $\text{rank } T = 2$  and, by [Theorem 3.24](#),  $\dim \ker T = 4 - 2 = 2$ . The free variables are  $x_2$  and  $x_4$ . Setting one of them to 1 and the other to 0 reads off the two basis vectors,

$$\ker T = \text{span}((-2, 1, 0, 0), (-1, 0, -1, 1)).$$

The image is the column space  $\text{span}((1, 2, 1), (0, 1, 1))$ , also of dimension 2, confirming  $4 = 2 + 2$ . Finally  $b = (1, 3, 2)$  is the fourth column  $Ae_4$ , so  $x_p = (0, 0, 0, 1)$  solves  $Ax = b$  and the complete solution set is  $x_p + \ker T$ .

◀

**Example 3.33 (Rank–nullity without matrices).** On  $\mathcal{P}_3(\mathbb{R})$ , which has dimension 4, let  $D$  be differentiation. Its kernel is the constants, so  $\dim \ker D = 1$ ; its image is all derivatives of cubics, namely  $\mathcal{P}_2(\mathbb{R})$ , of dimension 3. The theorem checks out:  $4 = 1 + 3$ . The same  $D$  is neither injective nor surjective as an operator on  $\mathcal{P}_3(\mathbb{R})$ , yet becomes surjective when viewed as a map  $\mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ . All four properties belong to a map with specified domain and codomain; in this comparison only the ambient codomain changes, preserving kernel and image dimension but changing surjectivity.

◀

**Remark 3.34 (Forward preview: operators in quantum mechanics).** This paragraph supplies motivation only: Lectures 7–9 define inner products, self-adjoint operators, and unitary/antiunitary transformations; Lecture 10 defines spectral projectors and the measurement postulates. In plain language, an observable is encoded by a self-adjoint linear operator, closed-system evolution uses a unitary operator that preserves lengths and overlaps, and a symmetry can be unitary or the conjugate-linear antiunitary kind. A spectral projector  $P_k$  keeps only the component associated with outcome  $k$ . For a normalized state vector, the later Born rule assigns that outcome probability  $\|P_k \psi\|^2$ ; after the outcome is known, the conditional update replaces the state by  $P_k \psi / \|P_k \psi\|$  when this norm is nonzero. That normalized replacement is not the observable's linear action. None of these later notions is needed to use kernel, image, or rank–nullity here. Kernels and images recur throughout: the kernel of  $\hat{H} - E I$  is the eigenspace at energy  $E$ , the object the spectral theorem (Lecture 10) will put at centre stage.

### Quotient spaces [self-study]

For a genuinely inhomogeneous system  $Ax = b$  with  $b \neq 0$ , a solution satisfies  $x_p \notin \ker T$ , so the solution set  $x_p + \ker T$  does not contain 0 and is not a subspace. For the homogeneous system  $b = 0$ , every particular solution lies in  $\ker T$  and the same expression is the zero coset  $\ker T$ , which is a subspace. The right language for all these cosets is the quotient space.

**Definition 3.35 (Quotient space).** Let  $U$  be a subspace of  $V$ . The *coset* of  $v \in V$  modulo  $U$  is

$$v + U = \{v + u : u \in U\}.$$

Two vectors are *equivalent*,  $v \sim v'$ , if  $v - v' \in U$ . The *quotient space*  $V/U$  is the set of all cosets, with operations

$$(v + U) + (w + U) = (v + w) + U, \quad \lambda(v + U) = (\lambda v) + U,$$

making it a vector space.

Both operations are defined through *representatives*, so they need checking. If  $v + U = v' + U$  and  $w + U = w' + U$ , then  $v - v' \in U$  and  $w - w' \in U$ , hence  $(v + w) - (v' + w') \in U$  and  $\lambda v - \lambda v' = \lambda(v - v') \in U$ : the answers  $(v + w) + U$  and  $(\lambda v) + U$  do not depend on which representatives were chosen. The vector-space axioms are then inherited from  $V$  one at a time, with  $0 + U = U$  as the zero vector.

Let  $V$  be finite-dimensional and let  $U \subseteq V$  be a subspace. Extend a basis  $u_1, \dots, u_m$  of  $U$  to a basis  $u_1, \dots, u_m, w_1, \dots, w_k$  of  $V$  (the extension theorem of Lecture 2). The cosets  $w_1 + U, \dots, w_k + U$  are then a basis of  $V/U$ . They span: writing  $v = \sum_i \alpha_i u_i + \sum_j \beta_j w_j$  and absorbing the  $u$ -part into  $U$  gives  $v + U = \sum_j \beta_j (w_j + U)$ . They are independent: if  $\sum_j \beta_j (w_j + U) = 0 + U$  then  $\sum_j \beta_j w_j \in U$ , so it is also a combination of the  $u_i$ , and independence of the full basis forces every  $\beta_j = 0$ . Counting lengths,

$$\dim(V/U) = \dim V - \dim U.$$

The same statement holds in arbitrary algebraic dimension, in the form  $\dim V = \dim U + \dim(V/U)$  as a sum of cardinals. That version is not available here: it needs the general basis-extension theorem, which rests on Zorn's lemma, and a notion of dimension for infinite-dimensional spaces, whereas Lecture 2 proves extension only in a finite-dimensional space and defines  $\dim$  only there.

The coset  $x_p + \ker T$  from the previous subsection is *exactly* an element of  $V/\ker T$ : the solution set of  $Ax = b$  is a single point in the quotient space, not in  $V$  itself. Different right-hand sides  $b$  correspond to different cosets—the quotient “forgets” what happens inside the kernel.

One word is needed before the next result, ahead of its proper treatment. Two vector spaces are *isomorphic*, written  $V \cong W$ , if some bijective linear map  $V \rightarrow W$  exists; such a map is an *isomorphism*. That is all the term means below. Lecture 4 studies it in depth and proves that for finite-dimensional spaces the dimension alone decides it.

**Example 3.36 (Cosets in  $\mathbb{R}^3$ ).** Let  $U$  be the  $xy$ -plane:

$$U = \text{span}((1, 0, 0), (0, 1, 0)) = \{(x, y, 0) : x, y \in \mathbb{R}\} \subseteq \mathbb{R}^3.$$

The coset of the vector  $v = (a, b, c)$  is the horizontal plane through the point  $(a, b, c)$ :

$$v + U = \{(a + s, b + t, c) : s, t \in \mathbb{R}\} = \{(x, y, c) : x, y \in \mathbb{R}\}.$$

This plane is parallel to the plane  $U$  and lies at height  $c$  (Figure 3). Two vectors have the same coset exactly when their  $z$ -coordinates agree. So an element of the quotient space

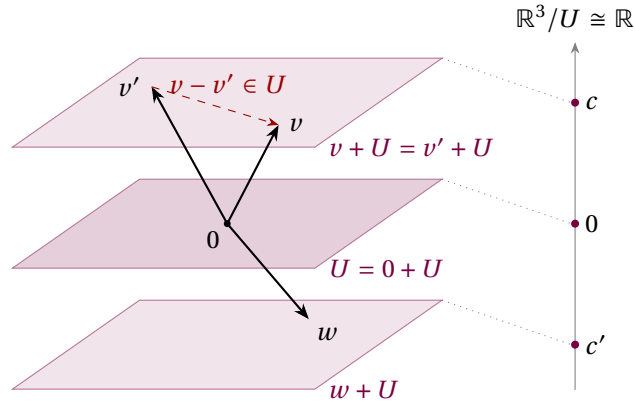


Figure 3: The cosets of the plane  $U$  in  $\mathbb{R}^3$  are the planes parallel to it. The vectors  $v$  and  $v'$  lie in the same coset, since  $v - v' \in U$ . The coset of  $w$  is a different plane. Each plane is one element of the quotient space  $\mathbb{R}^3/U$ , and a single number identifies it: the height of the plane.

$\mathbb{R}^3/U$  is identified by a single number  $c$ , and the operations on cosets become operations on these numbers: the sum of the planes at heights  $c$  and  $c'$  is the plane at height  $c + c'$ . Hence  $\mathbb{R}^3/U \cong \mathbb{R}$  (the  $z$ -axis), and the quotient space has dimension  $3 - 2 = 1$ .

The plane  $U$  is the kernel of the map  $T(x, y, z) = z$ , and the plane at height  $c$  is the solution set of the equation  $T(x, y, z) = c$ . Different right-hand sides  $c$  correspond to different cosets. ◀

**Theorem 3.37 (First Isomorphism Theorem).** For  $T \in \mathcal{L}(V, W)$ , the map  $\tilde{T}: V/\ker T \rightarrow W$  defined by

$$\tilde{T}(v + \ker T) = Tv$$

is well-defined, linear, and injective, with  $\text{im } \tilde{T} = \text{im } T$ . Hence

$$V/\ker T \cong \text{im } T,$$

without any finite-dimensional hypothesis. If  $V$  is finite-dimensional, the finite quotient formula recovers  $\dim V = \dim \ker T + \dim \text{im } T$  by comparing dimensions—an *alternative* view of Theorem 3.24, already proved directly above, rather than an independent proof of it.

*Proof. Well-defined.* If  $v + \ker T = v' + \ker T$  then  $v - v' \in \ker T$ , so  $T(v - v') = 0$  and  $Tv = Tv'$ .

*Linear.*  $\tilde{T}((v + \ker T) + (w + \ker T)) = T(v + w) = Tv + Tw = \tilde{T}(v + \ker T) + \tilde{T}(w + \ker T)$ , and similarly for scalars.

*Injective.*  $\tilde{T}(v + \ker T) = 0$  means  $Tv = 0$ , i.e.  $v \in \ker T$ , so  $v + \ker T = \ker T$  is the zero element of  $V/\ker T$ .

The image of  $\tilde{T}$  is the image of  $T$ . When  $V$  is finite-dimensional,  $\dim(V/\ker T) = \dim V - \dim \ker T$ , so comparing dimensions gives the stated rank–nullity equation. ■

The First Isomorphism Theorem is the canonical form of rank–nullity: it does not merely count dimensions but exhibits an explicit isomorphism.

**Remark 3.38 (Physics: gauge fields [self-study]).** Quotient spaces arise naturally in modern physics. For example, in electromagnetism the four-potential  $A_\mu$  is not unique: two potentials related by the transformation  $A_\mu \rightarrow A_\mu + \partial_\mu \chi$  produce the same field strength

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ . The “pure gauge” configurations (those of the form  $\partial_\mu \chi$ ) form a subspace of the space of all potentials, and the space of fields that physics “sees” as distinct is the quotient. A gauge-fixed description chooses one representative from each coset—just as choosing  $x_p$  picks one solution from each coset  $x_p + \ker T$ . You will meet gauge fields in the Electrodynamics course.

### Summary of main concepts

A linear map preserves linear combinations, hence superposition. It is fixed by its values on a basis, which may be chosen freely (**Theorem 3.5**). Its kernel  $\ker T$  and image  $\operatorname{im} T$  are subspaces;  $T$  is injective exactly when  $\ker T = \{0\}$  and surjective exactly when  $\operatorname{im} T = W$ . The rank–nullity theorem  $\dim V = \dim \ker T + \dim \operatorname{im} T$  ties the two together, yielding at a glance that maps to smaller spaces cannot be injective, maps to larger spaces cannot be surjective, and operators on a finite-dimensional space are injective if and only if surjective. In coordinates this is the bookkeeping of pivots and free variables behind every linear system.

- ▶ **Linear map**—preserves linear combinations (superposition); fixed by its values on a basis.
- ▶ **Kernel and image**—subspaces  $\ker T$  and  $\operatorname{im} T$ ; injective  $\iff \ker T = \{0\}$ , surjective  $\iff \operatorname{im} T = W$ .
- ▶ **Rank–nullity**— $\dim V = \dim \ker T + \dim \operatorname{im} T$ .
- ▶ **Operators: injective = surjective**—on a finite-dimensional space, injective  $\iff$  surjective.
- ▶ **Quotient space; First Isomorphism Theorem**— $V/\ker T \cong \operatorname{im} T$  (self-study); gauge equivalence in physics.

*Exercises for this lecture are in the companion sheet `exercises-3.pdf`.*